Swachh Bharat Mission (SBM) – Vision, Mission and Objectives

Introduction

Swachh Bharat Mission (SBM), also known as the Clean India Mission, was launched by Prime Minister Shri Narendra Modi on 2nd October 2014. The mission was started as a nationwide movement to achieve the vision of a Clean India and to pay tribute to Mahatma Gandhi on his 150th birth anniversary in 2019. The mission focuses on sanitation, hygiene, waste management and elimination of open defecation.

Vision of Swachh Bharat Mission

The vision of Swachh Bharat Mission is to make India:

- Clean and hygienic.
- Open Defecation Free (ODF).
- Environmentally sustainable.
- A country with improved sanitation facilities.
- A nation where people practice cleanliness as a habit.

The mission promotes sanitation, hygiene awareness and construction of toilets throughout the country.

Mission Approach of Swachh Bharat Mission

The mission follows a comprehensive approach that combines infrastructure development with behavioural change.

1. Behavioural Change (IEC)

IEC stands for Information, Education and Communication.

- Encourages people to adopt good sanitation habits.
- Creates awareness about cleanliness and toilet usage.
- Promotes Community-Led Total Sanitation (CLTS).

2. Community Participation

- Local communities participate in planning and monitoring.

- Schools, self-help groups and citizens are involved.
- Ensures long-term sustainability of sanitation facilities.

3. Financial Incentives

- Government provides financial support for toilet construction.
- Encourages construction of Individual Household Latrines (IHHL).
- Supports community sanitation complexes.

4. Sustainability through ODF+

The second phase focuses on maintaining ODF status by:

- Solid Waste Management.
- Liquid Waste Management.
- Plastic Waste Management.
- Greywater Management.

5. Technology and Monitoring

Technology is used for:

- GIS-based mapping.
- Geo-tagging of toilets.
- Mobile applications.
- Real-time monitoring of progress.

6. Capacity Building and Training

- Training is provided to sanitation workers and officials.
- Awareness programmes improve implementation efficiency.

7. Solid and Liquid Waste Management

- Door-to-door collection.
- Waste segregation at source.
- Waste-to-wealth projects.
- Scientific disposal of waste.

Objectives of Swachh Bharat Mission

1. Elimination of Open Defecation

The primary objective is to eliminate open defecation by:

- Constructing household toilets.
- Constructing community toilets.
- Constructing public toilets.

2. Improvement of Sanitation Infrastructure

- Construction of toilets.
- Development of sewage treatment facilities.
- Development of waste management systems.

3. Behavioural Change and Awareness

- Promotes hygiene awareness.
- Conducts cleanliness campaigns.
- Encourages community participation.

4. Solid and Liquid Waste Management

- Efficient waste collection.
- Waste segregation.
- Recycling and reuse.
- Safe disposal of waste.

5. Universal Sanitation Coverage

- Provides sanitation facilities in urban and rural areas.
- Ensures sanitation access for households, schools and public places.

6. Public Participation

- Encourages citizens to participate in cleanliness activities.
- Involves NGOs and private organizations.

7. Sustainability and Environmental Protection

- Promotes waste-to-energy projects.
- Encourages biogas generation.

- Supports scientific waste disposal methods.

SDG Targets of Swachh Bharat Mission

Swachh Bharat Mission supports the United Nations Sustainable Development Goals (SDGs), especially:

> **SDG 6 – Clean Water and Sanitation**
>
> The mission contributes to:
>
> - Safe sanitation facilities.
> - Improved hygiene practices.
> - Elimination of open defecation.
> - Better wastewater management.
> - Access to clean and safe sanitation for all citizens.

The mission also indirectly supports:

- **SDG 3** – Good Health and Well-being.
- **SDG 11** – Sustainable Cities and Communities.
- **SDG 13** – Climate Action through proper waste management.

Conclusion

Swachh Bharat Mission is one of India's largest cleanliness and sanitation programmes. It aims to make India clean, hygienic and open defecation free through sanitation infrastructure, behavioural change, community participation and scientific waste management. The mission also contributes significantly towards achieving Sustainable Development Goal 6 (Clean Water and Sanitation).

Explain the Salient Features of MSW Rules 2016

Introduction

Municipal Solid Waste (MSW) Rules, 2016 were introduced by the Government of India to ensure proper collection, segregation, transportation, processing and disposal of municipal solid waste. These rules aim to promote scientific waste management and reduce environmental pollution. The rules make citizens, local authorities, industries and institutions responsible for proper waste handling.

1. Applicability of the Rules

The MSW Rules 2016 apply to all waste generators and waste management authorities. The rules cover:

- Residential areas
- Commercial establishments
- Industrial establishments
- Institutional establishments
- Urban and rural areas
- Airports
- Railway stations
- Places of worship
- Special Economic Zones (SEZs)

This wide applicability ensures that waste generated from all sectors is managed properly.

2. Waste Segregation at Source

One of the most important features of the MSW Rules 2016 is segregation of waste at the source itself. Waste should be separated into three categories:

a) Biodegradable Waste (Wet Waste)

This includes:

- Food waste
- Vegetable peels
- Garden waste
- Organic waste

b) Non-Biodegradable Waste (Dry Waste)

This includes:

- Plastic
- Paper
- Glass
- Metal

c) Domestic Hazardous Waste

This includes:

- Batteries
- Electronic waste
- Chemical containers

The responsibility of segregation lies with households, offices and all waste generators. Proper segregation improves recycling and waste processing efficiency.

3. Collection and Transportation of Waste

The rules require local bodies to ensure proper collection and transportation of waste. Important provisions are:

- Door-to-door collection must be provided.
- Waste should be collected regularly.
- Transportation should be done using covered vehicles.
- Covered vehicles prevent littering and spillage during transport.

Proper collection and transportation help maintain cleanliness and reduce environmental pollution.

5. Role of Waste Generators

The rules assign responsibilities to waste generators. Their duties include:

- Segregating waste at source.
- Storing waste separately.
- Handing over waste to authorized collectors.
- Bulk waste generators such as hotels, offices and institutions must process their own waste.
- Plastic waste should be managed according to Plastic Waste Management Rules, 2016.

This helps reduce the burden on municipal authorities.

6. Extended Producer Responsibility (EPR)

Under EPR, manufacturers and brand owners are responsible for collecting and recycling waste generated from their products and packaging. Examples include:

- Plastic packaging
- Multi-layer packaging materials

This principle ensures that producers share responsibility for waste management and environmental protection.

7. Role of Urban Local Bodies (ULBs)

Urban Local Bodies play a major role in implementing the MSW Rules. Their responsibilities include:

- Ensuring scientific waste management.
- Providing collection and transportation services.
- Establishing waste treatment facilities.
- Conducting public awareness programmes.
- Imposing penalties for rule violations.

Thus, ULBs act as the primary implementing agencies for waste management.

8. Prohibition of Open Burning of Waste

The rules strictly prohibit open burning of municipal solid waste. Open burning causes:

- Air pollution
- Release of toxic gases
- Health hazards

Therefore, anyone violating this rule may be fined by local authorities.

9. Promotion of Decentralized Waste Processing

The rules encourage cities and towns to process waste close to the source of generation. This includes:

- Decentralized composting units
- Biogas plants

Benefits include:

- Reduced transportation cost
- Reduced landfill burden
- Faster waste treatment

Decentralized systems improve overall waste management efficiency.

10. Time-Bound Compliance Targets

The MSW Rules 2016 specify deadlines for the development of:

- Waste processing facilities
- Sanitary landfills
- Other waste management infrastructure

This ensures systematic and timely implementation of waste management practices.

4. Processing and Disposal of Waste

The MSW Rules 2016 promote scientific processing of waste instead of open dumping. The following methods are recommended:

- Composting and Vermicomposting for treating organic waste.
- Anaerobic Digestion and Bio-methanation for producing biogas from biodegradable waste.
- Material Recovery Facilities (MRFs) for recovering recyclable materials.
- Waste-to-Energy (WTE) Plants for treating non-recyclable waste.
- Sanitary Landfills for disposal of residual and inert waste.

Open dumping of waste is strictly prohibited.

Define Composting and Explain the Factors Affecting the Composting Process

Introduction

Composting is a biological process in which organic waste is decomposed by microorganisms such as bacteria, fungi and actinomycetes under controlled conditions to produce nutrient-rich compost. It is a natural, economical and eco-friendly method of treating organic waste. The compost produced can be used as a fertilizer and soil conditioner.

In municipal solid waste management, composting involves collecting the organic fraction of waste such as food scraps and garden waste and allowing it to decompose naturally through microbial activity under suitable conditions of moisture and aeration.

Factors Affecting the Composting Process

The efficiency of composting depends on several factors such as temperature, moisture content, carbon-to-nitrogen ratio, aeration and waste characteristics.

1. Temperature

Temperature plays an important role in microbial activity.

- The ideal temperature range for composting is 55°C to 71°C.
- Microorganisms become more active at higher temperatures.
- Increased microbial activity accelerates decomposition.
- High temperatures help destroy harmful pathogens that affect plants, animals and humans.

Importance: Proper temperature ensures faster decomposition and production of safe compost.

2. Moisture Content

Moisture is necessary for the growth and activity of microorganisms.

- The compost pile should feel like a wrung-out sponge.
- If the pile becomes too dry, microbial activity decreases and decomposition slows down.
- Excess moisture creates anaerobic conditions and causes foul odours.
- Proper moisture helps microorganisms break down organic matter efficiently.

Importance: Maintaining proper moisture improves the composting rate and compost quality.

3. Carbon-to-Nitrogen Ratio (C:N Ratio)

Microorganisms require both carbon and nitrogen for growth.

- Carbon acts as an energy source.
- Nitrogen is required for cell growth and reproduction.
- A high carbon-to-nitrogen ratio limits microbial growth.
- Improper C:N ratio slows the decomposition process.

Importance: A balanced C:N ratio ensures efficient microbial activity and rapid composting.

4. Aeration (Oxygen Supply)

Most composting processes are aerobic and require oxygen.

- Oxygen supports the growth of aerobic microorganisms.
- Proper aeration improves decomposition.

- Air can be supplied naturally by turning the compost pile or through forced aeration.
- Adequate aeration helps control odours.
- Lack of oxygen creates anaerobic conditions and foul smells.

Importance: Good aeration speeds up decomposition and prevents bad odours.

5. Texture and Size of Raw Materials

The physical characteristics of waste influence composting.

- Texture of raw materials affects air movement within the compost pile.
- Smaller particles provide more surface area for microbial action.
- Properly prepared waste decomposes more quickly.

Importance: Suitable particle size improves aeration and decomposition efficiency.

6. Volume of Waste

The amount of waste in the compost pile also affects composting.

- Larger piles retain heat more effectively.
- Waste volume influences temperature development within the pile.
- Proper pile size supports microbial growth and decomposition.

Importance: Adequate waste volume helps maintain optimum composting conditions.

Advantages of Composting

- Enhances soil structure, aeration and moisture retention.
- Provides slow-release nutrients such as nitrogen, phosphorus and potassium.
- Kills harmful bacteria, parasites and weed seeds due to high temperatures.
- Reduces methane emissions compared to landfill disposal.
- Diverts organic waste from landfills and reduces waste volume.

Conclusion

Composting is an eco-friendly biological process that converts organic waste into nutrient-rich compost. Factors such as temperature, moisture content, carbon-to-nitrogen ratio, aeration, texture of raw materials and waste volume significantly affect the composting process. Proper control of these factors ensures faster decomposition and production of high-quality compost.

Explain Anaerobic Digestion – Phases and Process

Introduction

Anaerobic digestion is a biological treatment process in which organic waste is broken down by microorganisms in the absence of oxygen. During this process, biodegradable waste such as food waste, manure, sewage sludge and crop residues are converted into useful products like biogas and digestate.

Anaerobic digestion is widely used in waste management because it reduces waste volume, produces renewable energy and provides nutrient-rich fertilizer.

Process of Anaerobic Digestion

The process of anaerobic digestion involves the following steps:

1. Waste Collection and Pre-Treatment

Organic wastes such as food waste, manure, sewage sludge and crop residues are first collected. Before the digestion process begins:

- Plastics, metals and other unwanted materials are removed.
- Moisture content is adjusted.
- Large waste particles are reduced in size to improve decomposition.

This step ensures efficient digestion of the waste.

2. Anaerobic Reactor

The prepared waste is then placed inside an airtight container called an anaerobic digester.

- Oxygen is completely absent inside the reactor.
- Suitable temperature and pH conditions are maintained.
- Microorganisms start breaking down the organic matter.

The decomposition takes place through four important biological phases.

Phases of Anaerobic Digestion

1. Hydrolysis

Hydrolysis is the first stage of anaerobic digestion. In this stage, complex organic substances such as carbohydrates, proteins and fats are broken down into simpler compounds.

- Carbohydrates are converted into sugars.
- Proteins are converted into amino acids.
- Fats are converted into fatty acids.

This stage prepares the waste for further decomposition by microorganisms.

2. Acidogenesis

In the second stage, microorganisms act on the simple compounds formed during hydrolysis. The sugars and amino acids are converted into:

- Organic acids
- Alcohols
- Hydrogen gas

These products serve as intermediate compounds for the next stage of digestion.

3. Acetogenesis

During acetogenesis, the organic acids produced in the previous stage are further broken down. They are converted into:

- Acetate
- Hydrogen
- Carbon dioxide

The acetate formed in this stage becomes the main raw material for methane production in the final stage.

4. Methanogenesis

Methanogenesis is the final and most important stage of anaerobic digestion. In this stage, methanogenic bacteria convert acetate and hydrogen into:

- Methane (CH_4)
- Carbon dioxide (CO_2)

The methane produced forms the major component of biogas.

End Products of Anaerobic Digestion

1. Biogas

Biogas is the main product obtained from anaerobic digestion. It contains:

- 50–70% Methane (CH_4)
- 30–50% Carbon dioxide (CO_2)

Biogas can be used for:

- Electricity generation
- Heating
- Cooking
- Vehicle fuel after purification

2. Digestate

Digestate is the residue left after digestion. It is rich in nutrients and can be used as an organic fertilizer to improve soil fertility and crop growth.

Benefits of Anaerobic Digestion

- Produces renewable energy in the form of biogas.
- Reduces the amount of organic waste sent to landfills.
- Provides nutrient-rich fertilizer.
- Reduces unpleasant odours and harmful pathogens.
- Helps reduce greenhouse gas emissions and environmental pollution.

Conclusion

Anaerobic digestion is an efficient method for treating organic waste. It takes place in the absence of oxygen and involves four stages: Hydrolysis, Acidogenesis, Acetogenesis and Methanogenesis. The process produces useful products such as biogas and digestate, making it an environmentally friendly and sustainable waste management technique.